Supplementary Information for: Certified symmetry discovery for polynomial and dictionary vector fields: confidence sets for Lie-algebra generators from noisy vector-field measurements

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Supplementary Note 1: Invariances of a learned model

The same machinery applies where F is a learned scalar function f and ξ generates an invariance: the Lie derivative becomes $\nabla f \cdot \xi$, and in place of Definition 1 we measure the departure from invariance by

$$\rho_\nu(\xi; f)^2 = \frac{\int (\nabla f \cdot \xi)^2 \, d\nu}{\int |\nabla f|^2 \, |\xi|^2 \, d\nu} \in [0, 1],$$

the root-mean-square cosine of the angle between the model's gradient and the generator, weighted by $|\nabla f|^2 |\xi|^2$. Like the defect of Definition 1, it is a ratio of norms of linear images of the coefficients. We fit degree-6 polynomial regressions, with features fixed in advance, to noisy samples (noise levels 0.01, 0.05 and 0.2; N from 1000 to 64,000; 300 replications per setting) of five planar targets: the rotation-invariant $r^2 - r^4/2$, three perturbations of it by εx^3 ($\varepsilon = 0.02$, 0.05 and 0.2, with rotation defects 0.025, 0.063 and 0.23 against the tolerance $\delta = 0.05$), and an anisotropic quartic with no invariance. The candidate generators are the linear vector fields on the plane, and ν is uniform on the square $[-1, 1]^2$. We then ask which invariances *of the target* may be declared (Supplementary Fig. S6). With the training inputs drawn uniformly from the whole square, both the threshold rule and SymCert recover the one-dimensional invariance once N is large enough: at noise

level 0.05 the threshold rule declares it from the smallest training set studied ($N = 1000$) and SymCert certifies it in 97% of replications at $N = 4000$ and in all of them from $N = 16{,}000$. For the perturbed targets, the invariance with defect 0.025 is declared by the threshold rule at every noise level once N is large enough (in at least 97% of replications from $N = 16{,}000$) and certified by SymCert at noise level 0.01 for every N and at 0.05 from $N = 16{,}000$; at noise level 0.2 SymCert certifies it in no replication up to $N = 16{,}000$ and in 8% (25 of 300) at $N = 64{,}000$. The two invariances with defects above the tolerance are declared by the threshold rule in at most 0.7% of replications and never by SymCert. Even with inputs covering the square the threshold rule is not free of false declarations: at noise level 0.2 and $N = 4000$ it declared a generator whose defect on the target exceeds δ in 3.3% of replications for the target with defect 0.025 and in 1.3% for the invariant target, whereas SymCert never certified falsely with these inputs. With the training inputs confined to the part of the square within a 120° sector ($|\arg x| < 60°$), the threshold rule declares invariances that the target does not have, at rates up to 0.72 at noise level 0.05 (Supplementary Fig. S6b) and up to 0.76 over the three noise levels, for both an invariant and a non-invariant target; SymCert never does.

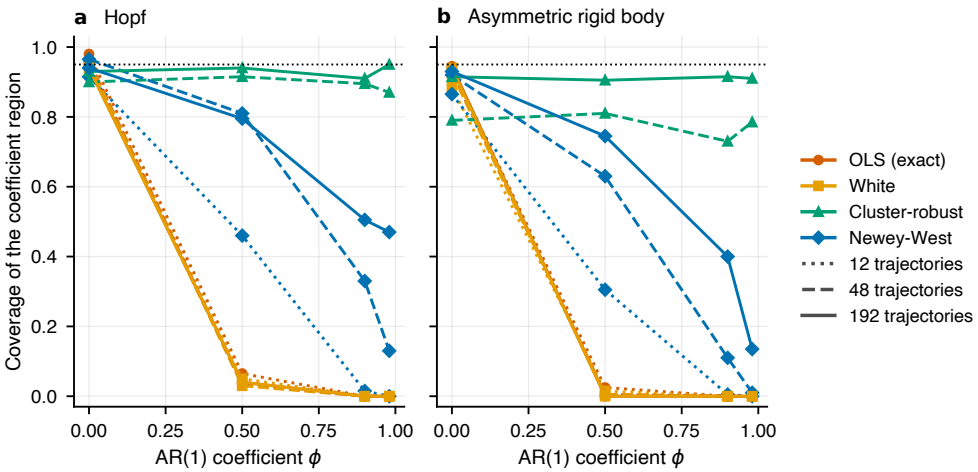
This is a second and distinct coverage mechanism. In the main text the *integral* was extrapolated: the Gram matrix computed on the data stood in for the one on the target domain, and κ measured the gap. Here the integral is exact and the *model* is extrapolated: outside the sector the fit is unconstrained, and it drifts toward functions that happen to look invariant. A diagnostic on the inputs alone can only warn that this may happen: the extrapolation factor of the main text, computed from the sector inputs for the polynomials of degree at most 6 in which the Lie derivatives lie, is between 2×10^4 and 3×10^4 (κ between 1.4×10^8 and 2.6×10^8) at every N studied, whereas for inputs covering the square the largest factor over settings is 2.2, 1.5, 1.2 and 1.1 at $N = 1000$, 4000, 16,000 and 64,000 (training inputs of the first replication of each setting). It cannot tell whether the drift has produced a spurious invariance, because that depends on the responses. SymCert can, because the same lack of constraint that lets the fit drift also inflates $\hat{\Sigma}$ in exactly those directions, and the certificate is then too wide to clear any useful tolerance. The region used here is the classical one, which is valid because the polynomial features are fixed in advance; for features learned on the same data it would not be (main text, "Learned features").

Supplementary Note 2: A case study on lynx–hare pelt records

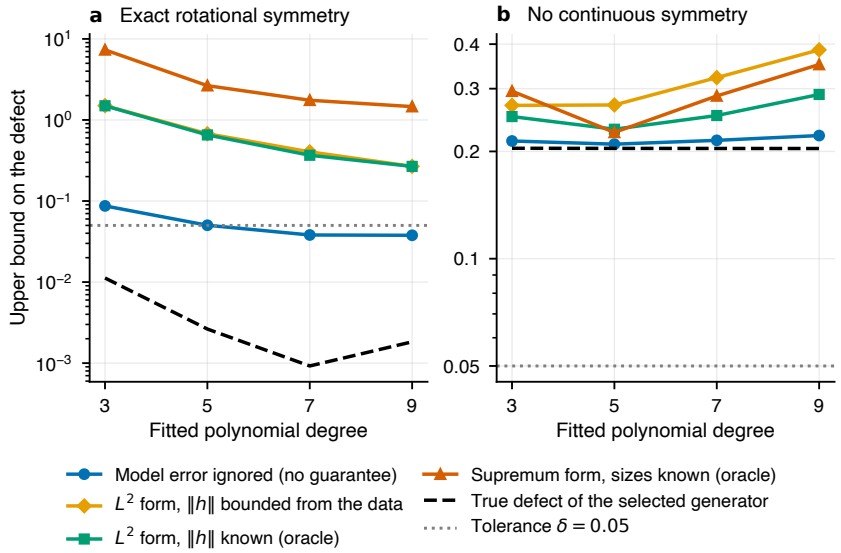
The Hudson's Bay Company lynx and hare pelt records for 1900–1920 are the canonical real data for two-dimensional predator–prey model discovery: 21 annual counts, from which central differences give 19 usable state–derivative pairs. Counts are expressed in units of the largest count (77.4 thousand pelts), and the target measure ν is uniform on $[-1, 1]^2$ in these units; the states lie in $[0.10, 1] \times [0.08, 0.77]$. Fitting a quadratic vector field ($Q = 12$) and running the pipeline with White's sandwich gives the plug-in defect spectrum (0.18, 0.33, 0.68, 1.17, 1.31, 1.40), an extrapolation factor $\sqrt{\kappa\kappa'} = 58$, and a smallest certified tolerance of 0.65: the data

support at best the statement that some generator's two bracket terms agree to within 65%. The target square is much larger than the region of the data; with the positive quadrant $[0, 1]^2$ as target the factor is 7.2, the plug-in spectrum begins $(0.20,\ 0.63,\ \dots)$ and the smallest certified tolerance is 0.84, so the conclusions below do not depend on this choice. With nineteen observations, no useful symmetry claim is supported. The plug-in spectrum nonetheless *looks* informative, and at a tolerance of 0.3, not an unreasonable choice, a threshold rule would report a one-dimensional symmetry algebra. We present this as a case study rather than as a validity claim: with 19 rows the sandwich region has no asymptotic justification, and derivatives differenced from noisy annual counts put error in the regressors, the departure the guarantee does not cover.

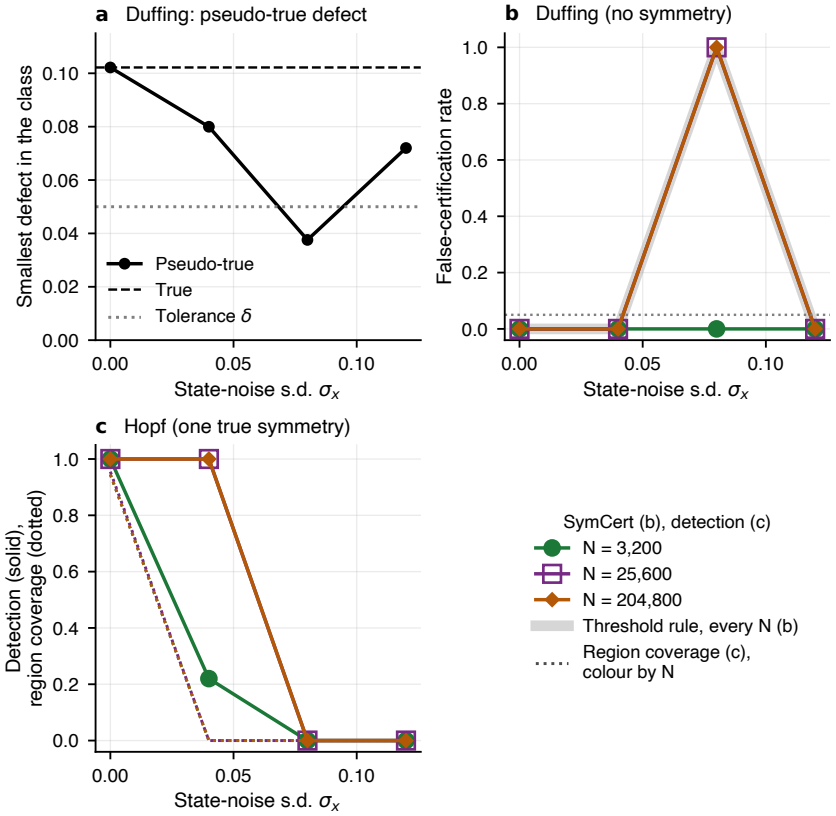
Supplementary Figures



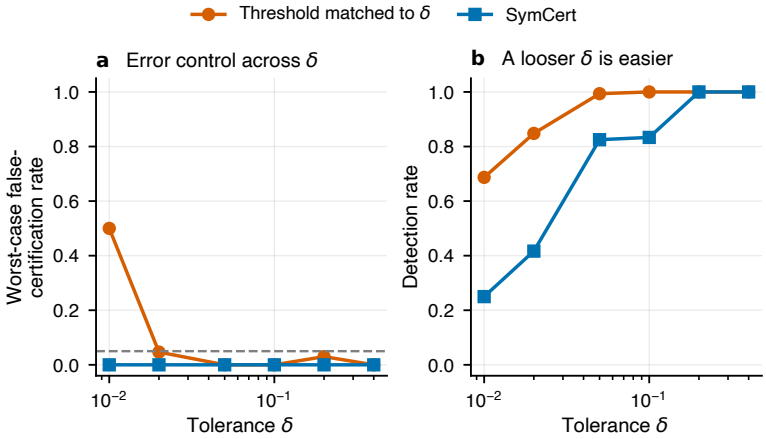
Supplementary Fig. S1 Serially correlated errors along trajectories. Coverage of the coefficient region at nominal 0.95 (horizontal black dotted line) when the errors on the right-hand side (noise level 0.05) follow an AR(1) process with coefficient ϕ along each trajectory, for the Hopf normal form (**a**) and the asymmetric rigid body (**b**), with 12 (dotted), 48 (dashed) or 192 (solid) trajectories and four regions: OLS (exact under independence), White, cluster-robust over trajectories (jackknife form) and Newey–West within trajectories. States are exact; each trajectory starts from a uniform initial condition in the box and runs for 100 steps of 0.01, only its part inside the box is kept (at most 100 points), and the AR(1) process restarts in each contiguous part. The cluster-robust region is not shown with 12 trajectories, fewer than the coefficients, where it leaves directions unidentified. 200 replications per point. Region and certificate coverages for every setting are in Supplementary Table S9, and no region gave a false certification in any setting; Table 3 of the main text summarises the case of 192 trajectories at $\phi = 0.9$ and 0.98.



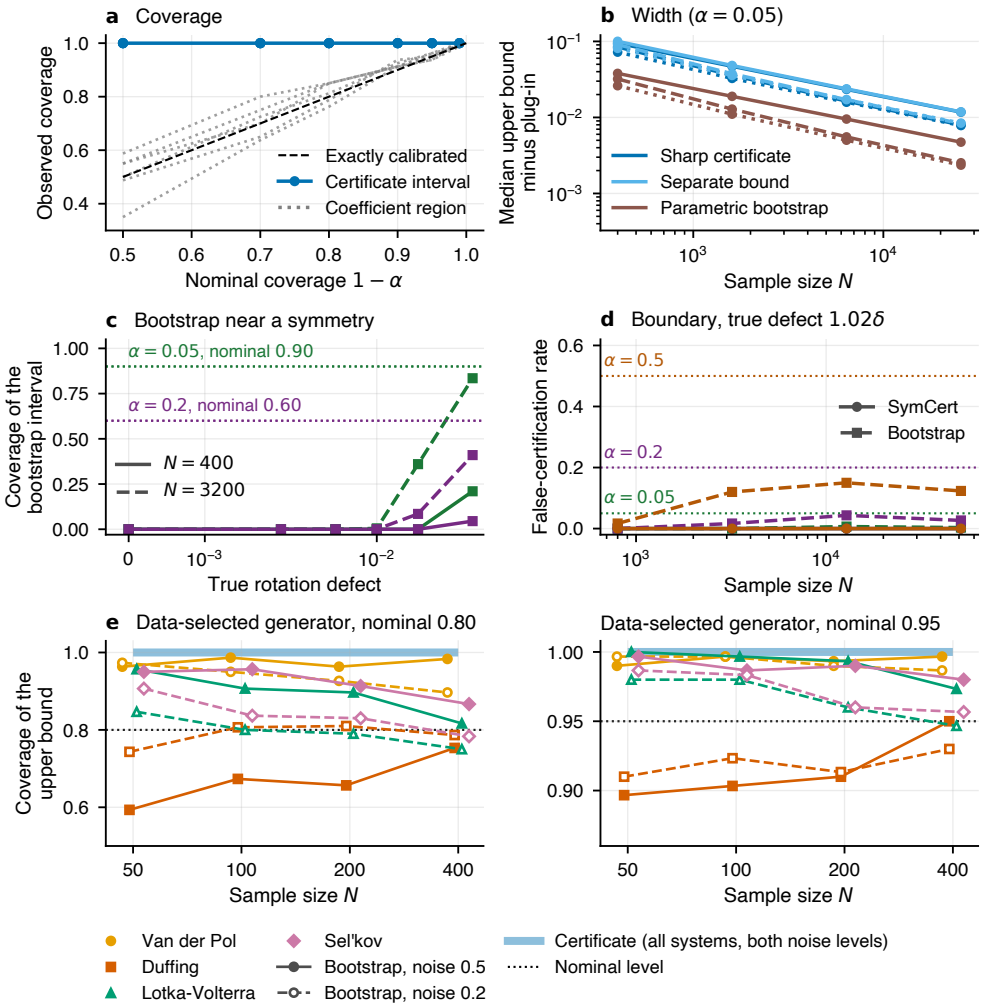
Supplementary Fig. S2 Certifying a system that is not in any polynomial class. Upper bounds on the defect of the selected generator (the minimal-defect direction of the fitted field) against the degree of the fitted polynomial class ($N = 40{,}000$, noise level 0.02; one data set per system and degree, not replicated), for the radial flow $r' = r(1 - r)$, $\varphi' = 1/(1 + r^2)$, which is exactly rotation invariant (**a**), and the damped pendulum, which has no continuous symmetry (**b**). Blue: model error ignored, which carries no guarantee. Green and orange: model error paid for in L^2 against a class four degrees larger (green) or in supremum norm (orange), with the size of the error *measured from the known truth* (oracle curves, unavailable in practice). Yellow: the L^2 form with the size of the omitted part bounded from the data. Both L^2 curves assume that the class four degrees larger contains the truth, which holds only approximately for these systems, so only the orange curve is a guaranteed bound, and only if its oracle sizes bound the true suprema; here they are maxima over 4×10^5 uniform points, which approach the suprema from below. Dashed: true defect of the selected generator, by Monte Carlo integration over the target measure (2×10^6 uniform points; standard errors below 2×10^{-4}) of the bracket of the true vector field, whose Jacobian is in closed form for the pendulum and by central differences (step 10^{-6}) for the radial flow; in **a** it is small but not zero because the selected direction differs slightly from the rotation, whose true defect is 0. Dotted: the tolerance $\delta = 0.05$.



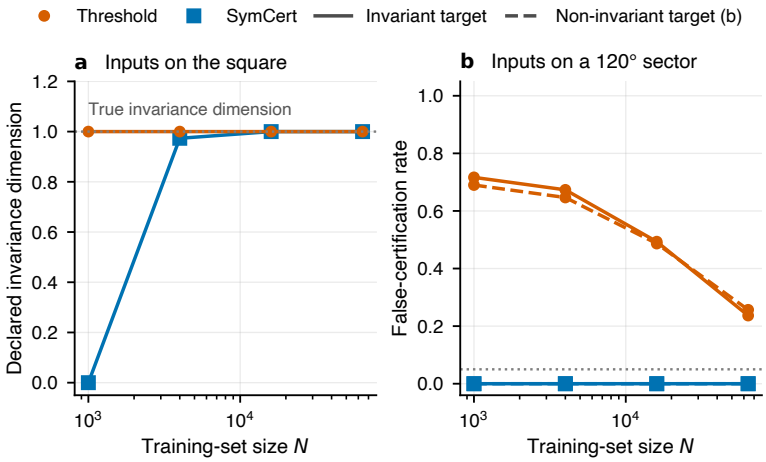
Supplementary Fig. S3 Noisy states can manufacture or hide symmetry. States observed with independent Gaussian noise of standard deviation σ_x (the target box has half-width 1), right-hand side measured separately with noise level 0.02; 200 replications per point. **a**, Smallest defect in the affine class of the pseudo-true field that least squares estimates (from a reference fit to 10^6 noisy states) for the hardening Duffing oscillator, against its true value and the tolerance. **b**, False-certification rate of SymCert at three sample sizes (colour and marker) and of the threshold rule (thick grey line, identical at every sample size): once the pseudo-true defect falls below the tolerance, both certify a false symmetry, the certificate as soon as N is large enough for it to be narrow. **c**, On the symmetric Hopf system state noise instead raises the pseudo-true defect of the rotation and, from $\sigma_x = 0.08$, destroys detection (solid); the coefficient region (thin dotted lines, same colours) misses the true coefficients whenever $\sigma_x > 0$.



Supplementary Fig. S4 Sweeping the tolerance. A smaller grid than the main calibration study: three symmetric systems (Hopf normal form, planar rotation, symmetric top) and three without symmetry (Van der Pol, Lotka–Volterra, asymmetric rigid body), $N = 400$ and 3200, noise levels 0.03 and 0.1, $\delta \in \{0.01, 0.02, 0.05, 0.1, 0.2, 0.4\}$, 300 replications each; it omits $N < 400$ and noise level 0.3, where detection is lower in the main study. **a**, Worst-case false-certification rate. SymCert's is zero at every tolerance; the threshold rule matched to the same δ exceeds the nominal level at the tightest tolerance and happens to stay below it at the others on this grid (on the full grid of the main calibration study its worst case is 0.63 at every threshold up to 0.1). **b**, Detection rate on the three symmetric systems: accepting a larger approximation makes symmetry easier to certify, from 0.25 at $\delta = 0.01$ to 1.00 at $\delta = 0.2$.



Supplementary Fig. S5 How conservative is the certificate, and what does a finite-sample guarantee cost? **a**, Observed against nominal coverage for the certificate interval (blue; all plug-in directions jointly) and the coefficient region (grey), one line per setting (three systems, $N = 400$, noise levels 0.03 and 0.1, 80 replications each). Pooled over the settings, the region's coverage is within 0.02 of nominal at every level; single lines scatter by up to 0.15, consistent with Monte Carlo error (standard error up to 0.056 per point). **b**, Median excess of the upper bound over the plug-in defect at $\alpha = 0.05$, for the sharp certificate U (dark blue), the separate bound U_{sep} of Proposition 2(i) of the main text (light blue) and the parametric bootstrap (brown), for the rotation generator of the broken-Hopf family at noise level 0.1 (200 replications), for true defects 0 (solid), 1.02δ (dashed) and 2δ (dotted). **c**, Coverage of the bootstrap interval $[q_\alpha, q_{1-\alpha}]$ (nominal $1 - 2\alpha$, dotted) as the true defect approaches zero (200 replications); the certificate's coverage is at least 0.93 throughout. **d**, False-certification rate at true defect 1.02δ for the rotation generator fixed in advance (300 replications); dotted: nominal levels. **e**, Coverage of the upper bound for the generator with the smallest plug-in defect, selected and bounded on the same data, on four systems without symmetry, at nominal 0.80 (left) and 0.95 (right), nominal levels dotted: bootstrap at noise levels 0.5 (solid, filled) and 0.2 (dashed, open); the certificate's coverage (thick light-blue line, lowest over systems and noise levels) is 1 in every setting; 300 replications per point, points offset slightly along N.



Supplementary Fig. S6 Invariances of a learned model (Supplementary Note 1). Both panels: noise level 0.05, degree-6 polynomial features, 300 replications per point. **a**, With training inputs covering the whole square $[-1, 1]^2$ on which the defect is defined, both rules eventually recover the one-dimensional rotational invariance of the target; certification needs about four times more data. **b**, With training inputs confined to the part of that square within a 120° sector, the fitted model is unconstrained elsewhere, and the threshold rule declares invariances that the target does not have. Solid lines: rotation-invariant target; dashed: target with no invariance. The two are almost indistinguishable, which is the point, since the threshold rule fails the same way whether or not there is anything to find. SymCert declines in both cases.

Supplementary Tables

system	unit factor c	defect of fixed generator		smallest defect in class	
		Euclidean	nondimensional	Euclidean	nondimensional
broken Hopf	0.01	0.0755	0.0644	0.0617	0.0525
	0.1	0.0753	0.0644	0.0616	0.0525
	0.5	0.0713	0.0644	0.0584	0.0525
	1	0.0644	0.0644	0.0525	0.0525
	2	0.0566	0.0644	0.0446	0.0525
	10	0.0511	0.0644	0.0375	0.0525
	100	0.0508	0.0644	0.0370	0.0525
Lotka–Volterra	0.01	0.2713	0.2713	0.0035	0.2713
	0.1	0.2713	0.2713	0.0351	0.2713
	0.5	0.2713	0.2713	0.1647	0.2713
	1	0.2713	0.2713	0.2713	0.2713
	2	0.2713	0.2713	0.1647	0.2713
	10	0.2713	0.2713	0.0351	0.2713
	100	0.2713	0.2713	0.0035	0.2713

Supplementary Table S1 The units of the state matter unless the metric is declared. One state coordinate is re-expressed in units c times larger ($x = \mathrm{diag}(c, 1)z$). The fixed generator is the rotation for the broken-Hopf system and the minimal-defect generator for Lotka–Volterra, transported to the new coordinates; "nondimensional" uses $W = \mathrm{diag}(1/s_i^2)$ with s_i the root-mean-square size of coordinate i over the target box, which coincides with the transported metric. Certification in the new units at $N = 12800$, noise level 0.05, judged by the nondimensional defect of the physical system (300 replications at $c = 0.01, 1, 100$, both systems, both metrics). With the Euclidean metric, Lotka–Volterra was certified in every replication at $c = 0.01$ and $c = 100$, and every certification was false; in every other case, and always with the nondimensional metric, nothing was certified.

system	N	noise	mean dim.	runs dim. ≥ 2	closure (median)	closure (max)	commutators certified
sphere flow, $\mathfrak{so}(3)$	800	0.01	3.00	100/100	0.001	0.003	1.00
sphere flow, $\mathfrak{so}(3)$	800	0.03	0.83	0/100	—	—	—
sphere flow, $\mathfrak{so}(3)$	3200	0.01	3.00	100/100	0.001	0.001	1.00
sphere flow, $\mathfrak{so}(3)$	3200	0.03	3.00	100/100	0.002	0.003	1.00
sphere flow, $\mathfrak{so}(3)$	12800	0.01	3.00	100/100	0.000	0.001	1.00
sphere flow, $\mathfrak{so}(3)$	12800	0.03	3.00	100/100	0.001	0.002	1.00
symmetric top	800	0.01	2.00	100/100	0.001	0.001	negligible
symmetric top	800	0.03	2.00	100/100	0.002	0.004	negligible
symmetric top	3200	0.01	2.00	100/100	0.000	0.000	negligible
symmetric top	3200	0.03	2.00	100/100	0.001	0.002	negligible
symmetric top	12800	0.01	2.00	100/100	0.000	0.000	negligible
symmetric top	12800	0.03	2.00	100/100	0.000	0.001	negligible
planar rotation	800	0.01	2.00	100/100	0.000	0.000	negligible
planar rotation	800	0.03	2.00	100/100	0.000	0.000	negligible
planar rotation	3200	0.01	2.00	100/100	0.000	0.000	negligible
planar rotation	3200	0.03	2.00	100/100	0.000	0.000	negligible
planar rotation	12800	0.01	2.00	100/100	0.000	0.000	negligible
planar rotation	12800	0.03	2.00	100/100	0.000	0.000	negligible

Supplementary Table S2 Closure of certified subspaces under the bracket. "Mean dim." is the mean dimension of the certified plug-in subspace. For each certified subspace of dimension at least two, the closure defect is the largest, over pairs of an $L^2(\nu)$-orthonormal basis, of the $L^2(\nu)$ norm of the part of the commutator outside the subspace relative to the size of its two terms (0 for a Lie subalgebra, at most $\sqrt{2}$). Affine generators are closed under the bracket, so each commutator is itself a candidate generator; the last column is the fraction of those runs in which every commutator larger than 5% of its two terms is certified at $\delta = 0.05$ ("negligible": all commutators smaller, as for an abelian algebra; "—": no subspace of dimension two or more was certified). 100 replications per row.

system	design	points	$\sqrt{\kappa\kappa'}$	true dim.	nullspace dim.	SymCert dim.
Van der Pol	single trajectory	3000	34.9	0	0	0
Van der Pol	8 trajectories	3200	9.7	0	0	0
Van der Pol	uniform design	3000	1.1	0	0	0
Hopf	single trajectory	3000	∞	1	1	0
Hopf	8 trajectories	3200	24.0	1	1	1
Hopf	uniform design	3000	1.1	1	1	1
Lorenz	single trajectory	3000	231.3	0	0	0
Lorenz	8 trajectories	3200	84.4	0	0	0
Lorenz	uniform design	3000	1.1	0	0	0

Supplementary Table S3 Trajectory data and the extrapolation factor. One realisation per row. Target measure uniform on $[-3, 3]^2$ (Van der Pol, degree-3 class), $[-1.5, 1.5]^2$ (Hopf, degree 3) and $[-20, 20]^3$ (Lorenz, degree 2); right-hand side measured with noise level 0.02, states exact. The single trajectory starts at 0.7 times the half-width in every coordinate ($(1, 1, 20)$ for Lorenz) and is sampled at 4000 times over $[0, 60]$ ($[0, 40]$ for Lorenz), the first 1000 discarded as transient; the eight trajectories start uniformly in the box and are sampled at 400 times over $[0, 8]$; the uniform design has 3000 points drawn from the target measure. The single Hopf trajectory converges onto the unit circle, an algebraic curve of low degree, so every field divisible by $x^2 + y^2 - 1$ in the space that contains every bracket vanishes on the data to within the eigenvalue threshold, and the factor is reported as infinite; on the Van der Pol limit cycle and the Lorenz attractor the data lie only near the zero sets of other nonzero fields of that space, and the factor is finite but large. Either way a defect measured on them says little about the target domain. The nullspace rule happens to give the right dimension on every row here, but on the single Hopf trajectory it does so without support from the data; SymCert declines to certify exactly there.

degree	Q	N	U (median)	pointwise (median)	Le Cam	U/Le Cam	pointwise/Le Cam
3	20	100	0.2880	0.4322	0.03536	8.1	12.2
3	20	200	0.1960	0.2423	0.02403	8.2	10.1
3	20	400	0.1332	0.1515	0.01655	8.0	9.2
3	20	800	0.0930	0.0993	0.01161	8.0	8.6
3	20	1600	0.0656	0.0684	0.00820	8.0	8.3
3	20	3200	0.0465	0.0464	0.00579	8.0	8.0
3	20	6400	0.0330	0.0320	0.00409	8.1	7.8
3	20	12800	0.0229	0.0222	0.00289	7.9	7.7
5	42	100	0.7941	5.0395	0.09140	8.7	55.1
5	42	200	0.5132	0.8830	0.04998	10.3	17.7
5	42	400	0.3429	0.4384	0.03234	10.6	13.6
5	42	800	0.2403	0.2683	0.02224	10.8	12.1
5	42	1600	0.1664	0.1701	0.01531	10.9	11.1
5	42	3200	0.1180	0.1136	0.01066	11.1	10.7
5	42	6400	0.0821	0.0771	0.00750	10.9	10.3
5	42	12800	0.0584	0.0537	0.00531	11.0	10.1
7	72	100	1.1233	∞	0.29852	3.8	∞
7	72	200	0.9649	∞	0.12489	7.7	∞
7	72	400	0.7326	1.6610	0.06745	10.9	24.6
7	72	800	0.5058	0.6841	0.04115	12.3	16.6
7	72	1600	0.3549	0.3896	0.02768	12.8	14.1
7	72	3200	0.2481	0.2422	0.01885	13.2	12.9
7	72	6400	0.1748	0.1584	0.01313	13.3	12.1
7	72	12800	0.1228	0.1075	0.00928	13.2	11.6

Supplementary Table S4 The gap to the Le Cam bound grows with the model class. The exactly symmetric Hopf system at noise level 0.1 fitted in polynomial classes of degree 3, 5 and 7; medians over 200 replications. U is the simultaneous certificate for the rotation generator, "pointwise" the Gaussian-concentration bound (valid because the rotation is fixed in advance), and Le Cam the bound of the main text at power 0.5 evaluated on each design. The pointwise bound is infinite (∞) for degree 7 at $N \leq 200$: its lower bound on the denominator of the defect reached zero in at least half of the replications. The median factors quoted in the main text are over $N = 800$ to 12,800, where every bound is finite and below 1.

k	N	rotation	min. true	lack-of-fit rejection		false certification					certified subspace dim.			
		defect/δ	defect	observed	theory	degree 3	gate	split	largest	simult.	gate	split	largest	simult.
5	800	0	0.0000	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	800	0.5	0.0240	0.09	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	800	1	0.0482	0.29	0.26	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	800	1.25	0.0604	0.38	0.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	800	1.5	0.0725	0.64	0.62	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	800	2	0.0969	0.89	0.91	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	800	3	0.1458	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	3200	0	0.0000	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.86	0.00	0.00	0.00
5	3200	0.5	0.0240	0.29	0.26	0.00	0.00	0.00	0.00	0.00	0.65	0.00	0.00	0.00
5	3200	1	0.0482	0.90	0.92	0.50	0.06	0.00	0.00	0.00	0.10	0.00	0.00	0.00
5	3200	1.25	0.0604	1.00	0.99	0.86	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	3200	1.5	0.0725	1.00	1.00	0.86	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	3200	2	0.0969	1.00	1.00	0.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	3200	3	0.1458	1.00	1.00	0.85	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	12800	0	0.0000	0.04	0.05	0.00	0.00	0.00	0.00	0.00	0.96	0.96	0.02	0.00
5	12800	0.5	0.0240	0.92	0.92	0.00	0.00	0.00	0.00	0.00	0.08	0.47	0.00	0.00
5	12800	1	0.0482	1.00	1.00	0.56	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	12800	1.25	0.0604	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	12800	1.5	0.0725	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	12800	2	0.0969	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	12800	3	0.1458	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	51200	0	0.0000	0.04	0.05	0.00	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00
5	51200	0.5	0.0240	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.97	0.09	0.97	0.94
5	51200	1	0.0482	1.00	1.00	0.54	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	51200	1.25	0.0604	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	51200	1.5	0.0725	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	51200	2	0.0969	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	51200	3	0.1458	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	0	0.0000	0.06	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	0.5	0.0245	0.05	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	1	0.0491	0.08	0.09	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	1.25	0.0614	0.11	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	1.5	0.0738	0.14	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	2	0.0985	0.25	0.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	800	3	0.1478	0.68	0.68	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	3200	0	0.0000	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.83	0.00	0.00	0.00
7	3200	0.5	0.0245	0.10	0.09	0.00	0.00	0.00	0.00	0.00	0.78	0.00	0.00	0.00
7	3200	1	0.0491	0.27	0.28	0.57	0.39	0.00	0.00	0.00	0.65	0.00	0.00	0.00
7	3200	1.25	0.0614	0.49	0.47	0.90	0.46	0.00	0.00	0.00	0.46	0.00	0.00	0.00
7	3200	1.5	0.0738	0.69	0.69	0.90	0.27	0.00	0.00	0.00	0.27	0.00	0.00	0.00
7	3200	2	0.0985	0.96	0.96	0.89	0.03	0.00	0.00	0.00	0.03	0.00	0.00	0.00
7	3200	3	0.1478	1.00	1.00	0.88	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	12800	0	0.0000	0.04	0.05	0.00	0.00	0.00	0.00	0.00	0.96	0.95	0.00	0.00
7	12800	0.5	0.0245	0.30	0.28	0.00	0.00	0.00	0.00	0.00	0.70	0.83	0.00	0.00
7	12800	1	0.0491	0.96	0.96	0.57	0.02	0.19	0.00	0.00	0.04	0.36	0.00	0.00
7	12800	1.25	0.0614	1.00	1.00	1.00	0.00	0.11	0.00	0.00	0.00	0.11	0.00	0.00
7	12800	1.5	0.0738	1.00	1.00	1.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00
7	12800	2	0.0985	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	12800	3	0.1478	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	51200	0	0.0000	0.06	0.05	0.00	0.00	0.00	0.00	0.00	0.94	0.94	0.00	0.00
7	51200	0.5	0.0245	0.94	0.96	0.00	0.00	0.00	0.00	0.00	0.06	0.41	0.00	0.00
7	51200	1	0.0491	1.00	1.00	0.50	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	51200	1.25	0.0614	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	51200	1.5	0.0738	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	51200	2	0.0985	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	51200	3	0.1478	1.00	1.00	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Supplementary Table S5 Choosing the model class from the data: all 56 settings. Hopf normal form plus $\varepsilon P_k(x)\partial_x$, with P_k the Legendre polynomial of degree k and ε set so that the true defect of the rotation is the stated multiple of $\delta = 0.05$; noise level 0.1, 300 replications per row. "Min. true defect" is the smallest true defect of any generator in the degree-k class; "theory" is the power of the lack-of-fit test from the noncentral F distribution. Procedures: certify in degree 3; the lack-of-fit gate; the gate applied on one half of the data with certification on the other (split); certify in degree k (largest); simultaneous over both classes. A false certification certifies a subspace containing a generator whose true defect exceeds δ; the last four columns give the mean dimension of the certified subspace. Where the rotation defect equals δ, the smallest true defect is 0.048 to 0.049, so a certified direction near the rotation can have a true defect just above δ; such certifications count as false.

family	ε	rotation defect	min. defect	true dim.	worst false cert.		dim.	U	refuted
					SymCert	threshold			
radial flow	0	0.000	0.000	1	0.000	0.003	1.00	0.008	0.00
radial flow	0.03	0.024	0.022	0	0.000	0.010	1.00	0.028	0.00
radial flow	0.06	0.048	0.044	0	0.000	0.063	0.47	0.050	0.00
radial flow	0.12	0.096	0.088	0	0.000	0.000	0.00	0.094	0.00
damped pendulum	0	—	0.204	0	0.000	0.000	0.00	0.213	1.00

Supplementary Table S6 Non-polynomial systems fitted with dictionaries that contain them. The planar radial flow $\dot{r} = r(1-r)$, $\dot{\varphi} = 1/(1+r^2)$ plus $\varepsilon x^2 \partial_x$ (exactly rotation invariant at $\varepsilon = 0$), on the unit disc, is fitted with quadratic polynomials and four radial functions; the damped pendulum $\dot{\theta} = \omega$, $\dot{\omega} = -\sin\theta - 0.1\,\omega$, on $[-1.5, 1.5]^2$, with a polynomial and trigonometric library. Affine candidate generators; the target measure is uniform on the domain. Worst false-certification rates are over $N = 200, 800, 3200$ and noise levels $0.03, 0.1, 0.3$, 300 replications each; "dim." (the mean dimension of the certified plug-in subspace) and the median upper bound U on the minimal-defect generator are at the largest sample size and smallest noise level; "refuted" is the largest fraction of replications in which the lower certificate excludes every δ-approximate symmetry.

n	P	Q	true dim.	median U, minimal-defect direction				detection rate				CPU-s / cert.
				N=200	N=800	N=3200	N=12800	N=200	N=800	N=3200	N=12800	
2	4	6	2	0.0143	0.0072	0.0036	0.0017	1.00	1.00	1.00	1.00	0.031
3	9	12	3	0.0222	0.0109	0.0054	0.0027	1.00	1.00	1.00	1.00	0.038
4	16	20	4	0.0272	0.0120	0.0068	0.0031	0.00	1.00	1.00	1.00	0.048
5	25	30	5	0.0392	0.0192	0.0096	0.0048	0.00	0.79	1.00	1.00	0.067
6	36	42	6	0.0420	0.0206	0.0107	0.0050	0.00	0.00	1.00	1.00	0.111

Supplementary Table S7 Scaling in the state dimension. Linear systems built from rotation blocks, whose symmetry algebra is the commutant of the coefficient matrix and has dimension n; noise level 0.05, 200 replications per cell. The first block is the median certified upper bound U on the defect of the plug-in direction with the smallest defect; the detection rate is the fraction of replications in which the dimension of the certified subspace is at least n. The candidate search space $P = n^2$ grows quadratically with no cost to validity; the width grows, but because P and Q grow together this design does not separate their effects. False certification was zero in all 20 cells (five dimensions, four sample sizes). CPU time is per certificate of the plug-in subspace, on one core.

degree (Q)	region	plug-in	U	L	certified subspace dim.	$\sqrt{\kappa\kappa'}$
1 (6)	OLS (exact)	0.025	0.040	0.010	2	1.02
1 (6)	White	0.025	0.042	0.008	2	1.02
1 (6)	cluster-robust	0.025	0.074	0.000	0	1.02
1 (6)	Newey–West	0.025	0.045	0.005	2	1.02
2 (12)	OLS (exact)	0.036	0.058	0.015	1	1.48
2 (12)	White	0.036	0.063	0.010	1	1.48
2 (12)	cluster-robust	0.036	0.107	0.001	1	1.48
2 (12)	Newey–West	0.036	0.069	0.006	1	1.48
3 (20)	OLS (exact)	0.037	0.064	0.012	0	2.69
3 (20)	White	0.037	0.070	0.008	0	2.69
3 (20)	cluster-robust	0.037	0.206	0.004	0	2.69
3 (20)	Newey–West	0.037	0.076	0.003	0	2.69
1 (6), window 37 h	cluster-robust	0.022	0.067	0.000	1	1.02
1 (6), window 49 h	cluster-robust	0.025	0.074	0.000	0	1.02
1 (6), window 73 h	cluster-robust	0.023	0.069	0.000	1	1.02
1 (6), summer 2016	Newey–West	0.096	0.200	0.000	1	1.07
1 (6), summer 2017	Newey–West	0.044	0.188	0.000	0	1.03
1 (6), summer 2018	Newey–West	0.032	0.061	0.003	2	1.02
1 (6), summer 2019	Newey–West	0.049	0.076	0.022	2	1.04
1 (6), summer 2018	cluster-robust	0.032	0.117	—	—	1.02
1 (6), summer 2019	cluster-robust	0.049	0.109	—	—	1.04
1–3, simultaneous	cluster-robust	—	0.254	—	—	—
1–3, simultaneous	Newey–West	—	0.079	—	—	—

Supplementary Table S8 Rotation invariance of near-inertial ocean currents. Hourly velocities of 31 drogued drifters of the Global Drifter Program (North Pacific, 30–45°N, 180–140°W, June–September 2016–2019; 46,733 state–derivative pairs), near-inertial band isolated by removing a running mean (49 hours unless stated), time in units of the local inertial period over 2π and velocity in units of its root-mean-square. U and L bound the defect of the rotation generator, fixed in advance by the physics; the target measure is the standard Gaussian in the normalised velocity. The cluster-robust region treats each drifter as a cluster; the Newey–West region allows serial correlation within each drifter's record. The certified subspace dimension is that of the certified plug-in subspace at $\delta = 0.05$. In the degree-1 class every fitted field $Ax + b$ commutes with the dilation about its fixed point and with itself, so two plug-in defects are zero whatever the data. These generators come first in the nested plug-in subspaces, and in the degree-1 rows the column counts how many of them the region is narrow enough to certify (0, 1 or 2 here). The column is therefore not evidence about the rotation, which is judged by U and L, and where it is positive while $U > \delta$ the certified subspace does not contain the rotation. For single summers the cluster-robust region is unidentified in 2016 and 2017, which have fewer drifters (3 and 4) than coefficients (6); for 2018 (10 drifters) and 2019 (15 drifters) it gives $U = 0.117$ and 0.109 (L and the certified subspace dimension are not reported for these two rows); the Newey–West rows cover all four summers. The last rows bound the rotation defect simultaneously over the classes of degree 1 to 3, assuming the largest contains the truth. Velocities carry measurement error (median standard error 0.014 $\mathrm{m\,s^{-1}}$ against a root-mean-square of 0.109 $\mathrm{m\,s^{-1}}$), which the guarantee does not cover.

system	ϕ	trajectories	region coverage				certificate coverage			
			OLS	White	cluster	N–W	OLS	White	cluster	N–W
Hopf ($Q = 20$)	0	12	0.980	0.955	—	0.915	1.000	1.000	—	1.000
	0	48	0.960	0.955	0.900	0.965	1.000	1.000	1.000	1.000
	0	192	0.940	0.940	0.930	0.940	1.000	1.000	1.000	1.000
	0.5	12	0.065	0.050	—	0.460	0.975	0.960	—	1.000
	0.5	48	0.035	0.030	0.915	0.810	0.975	0.970	1.000	1.000
	0.5	192	0.040	0.040	0.940	0.795	0.990	0.990	1.000	1.000
	0.9	12	0.000	0.000	—	0.015	0.190	0.140	—	0.980
	0.9	48	0.000	0.000	0.895	0.330	0.145	0.135	1.000	0.995
	0.9	192	0.000	0.000	0.910	0.505	0.155	0.150	1.000	1.000
	0.98	12	0.000	0.000	—	0.000	0.020	0.010	—	0.795
	0.98	48	0.000	0.000	0.870	0.130	0.025	0.010	1.000	0.990
	0.98	192	0.000	0.000	0.950	0.470	0.010	0.010	1.000	1.000
asymmetric rigid body ($Q = 30$)	0	12	0.940	0.885	—	0.865	1.000	1.000	—	1.000
	0	48	0.930	0.925	0.790	0.920	1.000	1.000	1.000	1.000
	0	192	0.945	0.940	0.915	0.930	1.000	1.000	1.000	1.000
	0.5	12	0.025	0.015	—	0.305	1.000	0.995	—	1.000
	0.5	48	0.005	0.005	0.810	0.630	0.995	0.995	1.000	1.000
	0.5	192	0.000	0.000	0.905	0.745	0.995	0.995	1.000	1.000
	0.9	12	0.000	0.000	—	0.005	0.265	0.220	—	1.000
	0.9	48	0.000	0.000	0.730	0.110	0.235	0.220	1.000	1.000
	0.9	192	0.000	0.000	0.915	0.400	0.240	0.225	1.000	1.000
	0.98	12	0.000	0.000	—	0.000	0.015	0.010	—	0.910
	0.98	48	0.000	0.000	0.785	0.010	0.015	0.015	1.000	1.000
	0.98	192	0.000	0.000	0.910	0.135	0.015	0.010	1.000	1.000

Supplementary Table S9 Serially correlated errors along trajectories. Errors on the right-hand side follow a stationary AR(1) process with coefficient ϕ along each trajectory, with exact states and noise level 0.05; each trajectory starts from a uniform initial condition in the box and runs for 100 steps of 0.01, only its part inside the box is kept, and the process restarts in each contiguous part. Regions: OLS (exact under independence), White, cluster-robust over trajectories (jackknife form) and Newey–West (N–W) within trajectories; nominal coverage 0.95, 200 replications per row. Region coverage is the fraction of replications in which the coefficient region contains the true coefficients; certificate coverage the fraction in which the interval contains the true defect of every plug-in direction. "—": with fewer trajectories than coefficients the cluster-robust region is not identified and certifies nothing. No region gave a false certification in any setting.